

B.I.R.D.S — User Manual

Broad Igneous Rocks — Diagrams & Statistics · Version 0.1.0

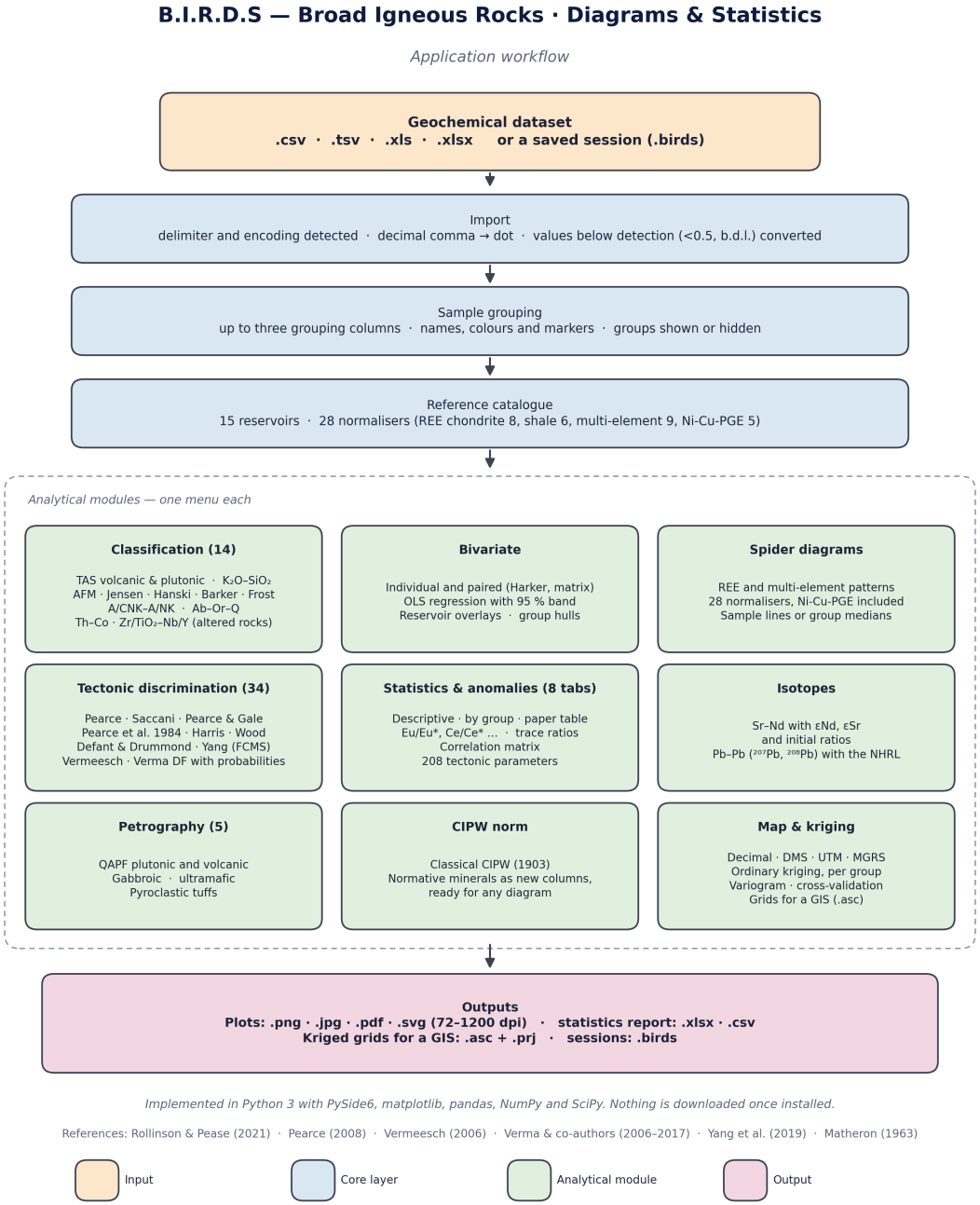


Figure 1. B.I.R.D.S application workflow.

1. About B.I.R.D.S

B.I.R.D.S (Broad Igneous Rocks — Diagrams & Statistics) is a desktop application for plotting and analysing whole-rock geochemical data of igneous rocks. It bundles 62 plotting templates, a spider-diagram engine with 28 published normalisers (REE chondrites and shales, multi-element references and Ni-Cu-PGE metal references), a discrimination module with 42 tectonic diagrams, and a statistics dashboard with REE-anomaly indices, a correlation-matrix heat-map and 256 tectonic discrimination parameters. A map view plots every sample at its longitude and latitude and kriges any numeric column across them, with the uncertainty of the result, and exports the surface as grids a GIS can open.

Every field boundary is transcribed, measured or recomputed from the original publication rather than redrawn by eye. docs/DIAGRAM_PROVENANCE.md records which of the three it was, diagram by diagram, together with the handful whose sources do not permit it and what each would need.

The software is built on top of the open scientific Python stack (PySide6, matplotlib, pandas, NumPy, SciPy) and runs identically on Windows, macOS and Linux. No prior programming experience is required to use it.

Citation

If you use B.I.R.D.S in a publication, please cite the underlying references printed in each plot's subtitle, plus the software itself: Torres Sánchez, D. (2026). B.I.R.D.S — Broad Igneous Rocks: Diagrams & Statistics (v0.1.0).

2. System requirements

- Operating system: Windows 10 or later, macOS 11 (Big Sur) or later, or any modern Linux distribution.
- Memory: at least 4 GB RAM (8 GB recommended for large datasets).
- Disk space: ~500 MB including Python + libraries.
- Python 3.9 or later (3.11 recommended).
- An internet connection is needed only for the initial library installation. Once installed, nothing in B.I.R.D.S uses the network.
- Installed as an application — BIRDS.app on macOS, or through the Windows installer — B.I.R.D.S needs no Python at all, and about 400 MB of disk.

3. Installation — Windows

3.1 Install Python

- Open your web browser and go to https://www.python.org/downloads/windows/
- Click the yellow “Download Python 3.x.x” button (3.11 or 3.12 are good choices).

• Run the downloaded installer. On the first screen, tick “Add python.exe to PATH” (this is important).

• Click “Install Now” and wait until it finishes. Click “Close”.

3.2 Download B.I.R.D.S

• Open the BIRDS distribution (e.g. a ZIP file or a folder you received).

• Extract it to a memorable location — for instance C:\Users\<your-name>\BIRDS

3.3 Install the required libraries

• Open the Start menu, type cmd and press Enter to open the Command Prompt.

• Navigate to the BIRDS folder you just extracted, for example:

```
cd C:\Users\<your-name>\BIRDS
```

• Install the required Python libraries by typing:

```
pip install -r requirements.txt
```

• Wait a few minutes while pip downloads PySide6, matplotlib, pandas, NumPy, SciPy and openpyxl.

3.4 Launch the application

• From the same Command Prompt window, type:

```
python main.py
```

• The B.I.R.D.S welcome screen should appear within a few seconds.

Note. Tip: to create a desktop shortcut, right-click an empty area of the desktop → New → Shortcut, paste the full path “C:\Python311\python.exe C:\Users\<your-name>\BIRDS\main.py” and name it B.I.R.D.S.

3.5 The installer

B.I.R.D.S can also be packaged as a Windows installer. The installed program needs no Python, has a Start-menu entry, and registers .birds sessions: they carry the session icon and open with a double-click. The installer is built on a Windows computer from the source, with PyInstaller and Inno Setup 6, following packaging/README.md; check the built program with BIRDS.exe --self-test before sharing it.

4. Installation — macOS

4.1 Install Python

• macOS comes with an old Python by default. Install a modern version from https://www.python.org/downloads/macos/

• Download the universal2 .pkg installer for Python 3.11 or 3.12.

• Run the installer and follow the on-screen instructions.

- Alternative: if you already use Homebrew, run `brew install python@3.11`.

4.2 Download B.I.R.D.S

- Extract the BIRDS folder to ~/Documents/BIRDS or any other convenient location.

4.3 Install the libraries

- Open Finder → Applications → Utilities → Terminal.
- Navigate to the BIRDS folder, e.g.:

```
cd ~/Documents/BIRDS
```

- Install the dependencies with pip:

```
python3 -m pip install --user -r requirements.txt
```

4.4 Launch the application

```
python3 main.py
```

> **Note.** On Apple-silicon Macs (M1/M2/M3) make sure you use the universal2 Python installer; PySide6 wheels are not yet available for some older Intel-only builds.

4.5 The application: BIRDS.app

Packaged as BIRDS.app, B.I.R.D.S runs without Python, opens from Launchpad or the Dock like any other program, and macOS knows its sessions: every .birds file carries the session icon and opens in B.I.R.D.S with a double-click.

- If you received BIRDS.app: drag it into Applications. The first time, right-click it and choose Open, then Open again — the app is not signed with an Apple developer certificate, so macOS asks once. After that it opens normally.
- To build it yourself, with the libraries of section 4.3 installed, add PyInstaller once and run the updater from the BIRDS folder:

```
python3 -m pip install pyinstaller
```

```
python3 packaging/update_mac_app.py
```

In about a minute it builds the app, checks it with its self-test and, only if that passes, puts it in Applications, replacing an older copy. Close B.I.R.D.S before running it.

4.6 Keeping the application up to date

BIRDS.app is a snapshot of the source as it was when built: changes to the source reach it only when it is rebuilt. Run python3 packaging/update_mac_app.py again — sessions, settings and the .birds association carry over. In the meantime python3 main.py always runs the source as it stands.

> **Note.** If .birds files still show a plain page right after the first install, relaunch the Finder: hold Option, right-click its icon in the Dock and choose Relaunch. macOS draws a new document icon in the background and shows the plain page until it is ready.

5. The B.I.R.D.S interface

When you launch B.I.R.D.S you are greeted by the welcome screen with the logo and a short tip. The menu bar at the top of the window holds eleven menus, from left to right:

- Data — sessions, data import, the data table, detection limits, plot style, pre-processing of the majors, and the export of reports and plots.
- Map — sample locations and kriging (section 12).
- Petrography — the IUGS modal diagrams: QAPF plutonic and volcanic (each with its own fields, numbered as the IUGS numbers them, with a key), gabbroic, ultramafic, Schmid's (1981) tuffs by glass, crystals and rock fragments, which the IUGS recommends, and Fisher's (1966) pyroclastic rocks by fragment size — blocks and bombs, lapilli, ash — from the Blocks_bombs_pct, Lapilli_pct and Ash_pct columns of the Petrography sheet. Fisher gives no percentages; his lines are drawn at thirds and are used as drawn.
- Norm — the classical CIPW norm.
- Classification diagrams — TAS, K2O–SiO2, AFM, Jensen, Frost and others, volcanic and plutonic.
- Bivariate diagrams — single and paired Harker-style plots.
- Spider diagrams — REE and multi-element patterns.
- Tectonic diagrams — 42 discrimination diagrams.
- Isotopes — Sr–Nd and Pb–Pb diagrams, and any isotope ratio, δ value or ε against SiO2.
- Statistics and Anomalies — descriptive statistics, REE anomalies, correlation matrix and tectonic discrimination parameters.
- References — the full bibliography.

A persistent toolbar above the plotting area shows a “← Home” button that always returns you to the welcome screen, no matter which window you are in.

Every control in the left-hand panel redraws the diagram as soon as you change it — log scales, the legend, fonts and axis titles, the regression line, the isotope reservoirs, the map's options. A text field applies when you press Enter or leave it, and a burst of changes, such as holding a spin box, is drawn once. Update plot (Ctrl+R or F5) is still there to redraw the view on screen at any time.

6. Loading your data

6.1 Supported formats

- .csv (comma, semicolon, tab or pipe delimited)
- .tsv (tab delimited)
- .xls and .xlsx (Microsoft Excel)

B.I.R.D.S auto-detects the delimiter and encoding (UTF-8 / Latin-1 / Windows-1252) and converts European decimal commas to dots automatically. Comment lines beginning with “#” at the top of the file are skipped.

6.2 Expected column names

Use standard element symbols and oxide formulas. Examples:

```
SiO2, TiO2, Al2O3, FeOT, MnO, MgO, CaO, Na2O, K2O, P2O5,
La, Ce, Pr, Nd, Sm, Eu, Gd, Tb, Dy, Ho, Er, Tm, Yb, Lu, Y,
Rb, Sr, Cs, Ba, Th, U, Nb, Ta, Zr, Hf, Pb, V, Sc, Co, Ni
```

Matching is case-insensitive (SiO2, sio2 and SiO2 are equivalent). At least one numeric column is required.

The template in the templates folder, BIRDS_data_template.xlsx, has a Data sheet for the bulk analyses and optional Isotopes and Petrography sheets, joined to Data by Sample_Name. The Isotopes sheet opens with the same columns, in the same order, as Data has before its major elements — identification, location, source, geology and analysis — so a sample's context can be copied across in one block; then comes SiO2, to read the isotopes against silica, and the isotope systems. Where a column is filled on both sheets, the Data value is kept, except that a blank Sample_Label or Group fills in from Isotopes. Age is read with its Age_Unit (Ma, ka, Ga or yr) for the initial-ratio correction.

6.3 Import dialog

- Click Data → Import data (.xls / .csv)…
- Browse to your file and select it.
- For Excel files, pick the sheet you want to load.
- A preview of the first 20 rows confirms the columns were parsed correctly.
- Click “Import” to load the data into B.I.R.D.S.

6.4 Define groups (recommended)

- Open Data → Edit plot.
- Pick up to 3 grouping columns (for instance: Type, Locality, Age).
- B.I.R.D.S auto-fills the group names from the unique values in those columns — up to 25 groups, each with its own colour and no two with the same colour and symbol. You can edit each name, colour and symbol. With many groups, set the legend to Outside so it does not cover the plot.
- Click OK. From now on every plot will respect those group styles.

6.5 Adjusting the majors to 100 %

Data → Pre-process majors… recalculates the major oxides to 100 wt % — anhydrous by default, removing LOI, H2O and CO2 — and splits total iron between Fe2O3 and FeO after Middlemost (1989), Le Maitre (1976) or a fixed ratio. A preview shows the values before and after.

Applying keeps the analysed values and adds the adjusted ones as new columns with the suffix _adj: SiO2_adj, TiO2_adj, …, FeO_adj, Fe2O3_adj and FeOt_adj. An oxide that was not analysed stays empty in its _adj column rather than becoming 0. Running the dialog again replaces the _adj columns, always starting from the analysed values.

Every view that reads major elements — classification, tectonic, multi-element spider, the CIPW norm and the statistics — carries a Major elements panel with the switch Use majors adjusted to 100 %. It is one switch: ticking or unticking it in any view changes all of them, so no two views can disagree about which values they use. The panel says which values are in use and how they were adjusted. It turns on when the adjustment is applied, and it is saved with the session.

> **Note.** The bivariate view has no switch: it lists the analysed and the adjusted columns side by side, so each axis shows which one it is by its name (section 8).

6.6 The data table

Data → View data table… (Ctrl+T) shows the Data, Isotopes and Petrography sheets as loaded, for checking and correcting in place: edit a cell, add or remove rows and columns, undo and redo. Each row is headed by its sample name, and a click on a column header sorts by it — numbers in numeric order, blank cells last.

The Data sheet ends with green calculated columns, per sample: the Mg# (Mg#, Mg#_FeOt and, once the majors are adjusted, Mg#_adj), the trace-element ratios, the alteration indices CIA and AI_Ishikawa, and the REE and multi-element anomalies, pink where positive and blue where negative. They are the values the Statistics view shows (section 11), against the chondrite picked under Anomaly reference — one choice for both places. Hover over a header for what each one is.

- They are read-only and never saved into your data: they are worked out again whenever a value changes, so they always follow the table.
- A column of your own with the same name — a Mg# you computed yourself — is shown as yours instead.
- Untick Calculated columns to hide them.
- Export table… saves the sheet, with the calculated columns, as Excel or CSV: every row, in file order.

7. Classification diagrams

Open Classification diagrams → Volcanic (or → Plutonic). The window has two panels: controls on the left, the plot on the right.

Available diagrams (volcanic)

- TAS — Total Alkalis vs Silica (Le Bas et al. 1986).
- K2O vs SiO2 — Peccerillo & Taylor (1976).
- Th vs Co for altered arc rocks — Hastie et al. (2007).
- Zr/TiO2 vs Nb/Y for altered rocks — Winchester & Floyd (1977).
- Alkalic / Sub-alkalic boundary on TAS (Irvine & Baragar 1971).
- AFM — Tholeiitic vs Calc-alkaline (Irvine & Baragar 1971 + Rollinson 2021).
- Jensen cation plot — high-Mg discrimination.
- Hanski [Al2O3] vs [TiO2] for komatiites.

- High-Mg subdivision on TAS (Le Maitre et al. 2002).

Available diagrams (plutonic)

- TAS plutonic — Middlemost (1994).
- Normative An–Ab–Or — Barker (1979).
- Frost et al. (2001) — Fe-index, MALI, ASI, AI vs FSSI.
- A/CNK vs A/NK aluminium saturation — Maniar & Piccoli (1989).
- Ab–Or–Q haplogranite — Tuttle & Bowen (1958).

Diagrams drawn from the norm

Barker's An–Ab–Or, the haplogranite Ab–Or–Q and panel (d) of Frost, AI vs FSSI, plot normative minerals. They compute the CIPW norm themselves from the majors in use, the same norm as the Norm view and following the same switch, so there is no need to visit that view first. A dataset that brings its own norm, in columns named Q, Or, Ab, An, Ne, Lc and Kp, is plotted as it stands instead.

A sample lacking SiO2, Al2O3, MgO, CaO, Na2O, K2O or the iron as FeO or Fe2O3 gets no norm and is left off: computed anyway, its blanks would count as zeros — without Al2O3 a granite has no feldspar and sits on the quartz apex. A note on the diagram says how many samples were left off and why, and the Norm view lists them as incomplete. Iron given only as a total (FeOt or Fe2O3t) must be split first: run Data → Pre-process majors… and tick Use majors adjusted to 100 %.

Naming a rock from the TAS

Both TAS diagrams can name a composition as well as plot it. The naming regions are built from the polygons the diagram draws, so the name and the picture cannot disagree.

- Volcanic — Le Bas et al. (1986). Tephrite/basanite and trachyte/trachydacite are separated by normative olivine and quartz; without a CIPW norm the pair is returned whole rather than guessed at.
- Plutonic — Middlemost (1994), which renames the volcanic fields (gabbro for basalt, diorite for andesite, granite for rhyolite, foid gabbro for tephrite, and so on) and adds a quartz monzonite field between granite and syenite. It needs no normative input.

> **Note.** Both require the analysis recalculated to 100 % anhydrous, as the IUGS scheme does. A composition outside the published diagram is left unnamed rather than forced into the nearest field — on the plutonic TAS that includes quartzolite and the tawite/urtite/italite group, which Middlemost places outside the main sequence of plutonic rocks.

Use the controls on the left to edit axis labels, font, legend location, and to enable a convex-hull shaded range per group.

8. Bivariate diagrams

Two sub-modes are available from the menu:

- Individual — single bivariate plot. Pick any two numeric columns for X and Y.

• Paired — multi-panel Harker / matrix plot. Select one X and several Y columns (or vice-versa) for a near-square grid; or several of both for a full X × Y matrix.

Adjusted majors and Mg#

Once the majors have been adjusted (section 6.5), each _adj column is listed right after the oxide it adjusts, in blue, so SiO2 and SiO2_adj sit side by side. After MgO come the magnesium numbers, in green, 100·Mg/(Mg+Fe²■) in molar proportions:

• Mg# — with the measured FeO. Offered only when FeO and Fe2O3 were reported separately.

• Mg#_FeOt — with all iron as Fe²■, what most papers mean when only total iron was measured. The 100 % adjustment does not change it.

• Mg#_adj — of the adjusted majors, with Fe²■ the FeO of the iron partition chosen when adjusting.

They are the same definitions as the statistics table uses, and they can be used inside an expression too, such as Mg#_FeOt/100. A column of your own named Mg# takes precedence over the computed one.

Optional overlays

• Linear regression line per group or overall, with optional 95 % confidence band and printed equation + R² in the legend.

• Convex-hull shaded range per group.

• Reservoir overlays — tick reference compositions (Bulk CC, N-MORB, OIB, KLB-1, …) to add labelled stars at the corresponding (X, Y) coordinates. Each answers in the template's names: total iron as FeOt (or Fe2O3T), Ni and Cr in ppm from the NiO and Cr2O3 its paper prints, K, Ti and P from their oxides. A reservoir with nothing for an axis is left off that plot — MIX1G on Ni, say, since its paper gives no NiO.

Annotations and Identify sample

Both sub-modes have the Annotations panel: lines, arrows, text and Identify sample, which labels the sample nearest the click. In Paired each annotation belongs to the panel clicked in — the sample is looked up on that panel's X and Y — and stays with that pair of variables when the grid is rearranged; a pair taken off the grid keeps its annotations for when it comes back. A line or arrow needs both clicks on the same panel. Right-click removes the nearest annotation, and samples of a group switched off cannot be identified. Alt / Cmd + click selects a sample here as in every other view, and in Paired it is ringed on every panel. Likewise, in the views that show many diagrams — Classification, Tectonic, Petrography and Isotopes — each diagram keeps its own annotations: an arrow drawn on TAS stays on TAS.

9. Spider diagrams

Choose Spider diagrams → REE for rare-earth patterns, or → Multi-elements for incompatible-element patterns. The reference combo on the left lets you pick:

• 8 REE chondrite normalisers (Sun & McDonough 1989, McDonough & Sun 1995, Pourmand 2012, Barrat 2012, Boynton 1984, Anders & Grevesse 1989, Taylor & McLennan 1985, Wakita 1971).

• 6 REE shale composites (NASC, PAAS, MUQ, European Shale, World Shale Average, UCC).

• 9 multi-element references (Primitive Mantle ×2, CI Chondrite, N-MORB ×2, E-MORB, UCC, Bulk CC, NASC).

• 5 Ni-Cu-PGE metal references (primitive mantle after McDonough & Sun 1995, Becker et al. 2006 and Fischer-Gödde et al. 2011; CI chondrite after Fischer-Gödde et al. 2010; depleted mantle after Salters & Stracke 2004), drawn on their own axis — Ni, the platinum-group metals, Au and Cu in order of compatibility with sulphide liquid.

Note. Every Ni-Cu-PGE value has been checked page by page against its paper. The primitive mantle often cited as Barnes & Maier (1999) is McDonough & Sun's (1995), and is listed under their name.

Plot modes

• Individual sample lines (default).

• Group summary — median + min–max envelope per group.

Reservoir patterns can be overlaid as dashed reference lines. Major elements K, Ti and P are auto-converted from wt.% oxide to ppm element so they sit on the same vertical scale as the trace data. KLB-1, KR4003 and MIX1G are given as major elements only, have no pattern, and are offered in Bivariate instead.

10. Tectonic discriminators

Open Tectonic diagrams → Volcanic or → Plutonic. Inside the window, choose the composition (Mafic, Intermediate or Acid) and the specific discriminator from the second combo. 42 published discriminators are implemented, each with the boundaries of its own source. Each plot prints its citation as a subtitle, and the entry in the combo names the axes in the order the published figure draws them — y first, then x.

Mafic

• Pearce (2008) Th/Yb vs Nb/Yb — crustal-input proxy.

• Pearce (2008) TiO2/Yb vs Nb/Yb — melting-depth proxy. The y axis is TiO2 in wt.% over Yb in ppm, as published.

• Saccani (2015) Th_N vs Nb_N — MORB-OIB array against volcanic arc array, N-MORB normalised.

• Saccani (2015) Yb_N vs Dy_N — boninite against island-arc tholeiite, chondrite normalised.

• Saccani (2015) (Dy/Yb)_N vs (Ce/Yb)_N — N-MORB against G-MORB, chondrite normalised.

• Pearce & Gale (1977) Zr/Y vs Ti/Y — within-plate against plate-margin.

• Shervais (1982) / Vermeesch (2006a) Ti vs V.

• Pearce & Cann (1973) / Vermeesch (2006a) Zr vs Ti.

• Five Vermeesch (2006a) ternaries: Na–25Nb–Sr, Si/1000–Ti/40–Sr, 100Eu–500Lu–Sr, V–Ti/50–5Sc and 50Sm–Ti/50–V.

• Wood (1980) / Vermeesch (2006a) Th – Hf/3 – Ta ternary.

• Cabanis & Lecolle (1989) La/10 – Y/15 – Nb/8 ternary, with Y/15 at the apex.

• Pearce (1976) / Vermeesch (2006a) major-element DF, and Butler & Woronow (1986) / Vermeesch (2006a) Zr–Y–Sr–Ti DF.

• Agrawal et al. (2004), Verma et al. (2006) and Agrawal et al. (2008) — the group's three earlier basic-rock sets, five DF panels each over IAB, CRB, OIB and MORB: from the adjusted oxides themselves, from ten ln(oxide/SiO2), and from ln(La, Sm, Yb, Nb / Th). Their papers print no centroids, so they give scores and fields but no probabilities.

• Verma & Agrawal (2011) — basic and ultrabasic rocks, five triplet DF panels.

• Yang et al. (2019) ln(FeO/CaO) vs ln(MgO/SiO2), the FCMS plane, and FCKANTMS vs ln(SiO2/(CaO+Na2O+TiO2)).

• Li, Godang & Saputro (2025) Nb/Yb vs Th/Nb — nine settings, from N-MORB to alkaline arc; for mafic to felsic rocks, so it is listed under all three compositions. The menu entry names its axes y first; on the plot Th/Nb runs across.

• Godang, Saputro & Li (2024) Nb/Zr vs Th/Nb, as Li et al. (2025) Figure 5 draws it — Nb/Zr sorts the mantle source (enriched above the primitive mantle's 0.0627, slightly enriched down to 0.04, depleted below) and Th/Nb the setting: the non-subduction array, the intracontinental rift, continental extension or alkaline arc, and the continental and island arcs either side of a diagonal from PM. Also under all three compositions.

Intermediate

• Defant & Drummond (1990) Sr/Y vs Y — adakite.

• Defant & Drummond (1993) Sr/Y vs Y — the adakite, island-arc ADR and Juan de Fuca Ridge fields drawn, measured on their Figure 2 (frame to Sr/Y 60).

• Drummond & Defant (1990, JGR) (La/Yb)N vs YbN — Archean TTG, post-Archean granitic rocks and MORB, measured on their Figure 10 and normalised to Wakita et al. (1971), the chondrite the paper names; the dotted line is the adakite limit Yb ≤ 1.9 ppm of Defant & Drummond's Nature paper.

• Drummond & Defant (1990, JGR) Sr/Y vs Y — high-Al TTD against the island-arc ADR fractionation suite, Figure 11; the ADR field is the envelope of the stipple the figure draws it with. Both Drummond & Defant figures are also under Acid, volcanic and plutonic.

• Martin (1986) (La/Yb)N vs YbN — Archean TTG against post-Archean granitoids, both fields measured on his Figure 1. Martin does not name his chondrite; Boynton (1984) is used. Also under Acid.

• Verma & Verma (2013) — three five-panel DF sets: majors, immobile majors + traces, and immobile traces.

• Li et al. (2025) Nb/Yb vs Th/Nb and Godang et al. (2024) Nb/Zr vs Th/Nb (see Mafic).

Acid (granitoids)

• Pearce et al. (1984) Ta vs Yb, Nb vs Y, Rb vs (Yb+Ta) and Rb vs (Y+Nb).

- Harris et al. (1986) Rb/30 – Hf – Ta×3 ternary, with Rb/30 at the apex.
- Drummond & Defant (1990, JGR) (La/Yb)N vs YbN and Sr/Y vs Y (see Intermediate).
- Martin (1986) (La/Yb)N vs YbN (see Intermediate).
- Verma, Pandarinath & Rivera-Gómez (2012) — acid rocks, five triplet DF panels.
- Verma et al. (2013) — three further five-panel DF sets: majors, immobile majors + traces, and immobile traces.
- Li et al. (2025) Nb/Yb vs Th/Nb and Godang et al. (2024) Nb/Zr vs Th/Nb (see Mafic).

Note. The Verma sets assume the rock type has already been fixed, with SINCLAS or IgRoCS. Nothing in B.I.R.D.S enforces that: fed a basalt, the intermediate-magma set will still answer, and confidently. Choose the set that matches your rocks.

11. Statistics and Anomalies

Click Statistics and Anomalies → Open dashboard to compute every available statistic in one click. A “Refresh” button and a chondrite-reference selector live at the top of the window. Eight tabs hold the results:

- Descriptive — count, mean, std, percentiles, median, MAD, %RSD, skewness and kurtosis for every numeric column, and for every trace ratio, alteration index and anomaly of the two per-sample tabs.
- By group — the same statistics broken down by your grouping column(s), ratios and anomalies included.
- Paper table — a publication-ready mean ± 1SD (n) per group, with the Mg# at the foot of the major elements and each adjusted oxide beside the analysed one. The trace ratios, alteration indices and anomalies are listed under their own headings, unticked, to add as needed.
- Anomalies & ratios — Eu/Eu*, Ce/Ce*, Pr/Pr*, Gd/Gd*, Dy/Dy*, Y/Y*, Y/Ho (raw), the primitive-mantle Ta/Ta*, Nb/Nb* and Ti/Ti*, Verma's Nb/Nb* and Ta/Ta*, and the normalised (La/Yb)_N, (La/Sm)_N, (Gd/Yb)_N, (Sm/Yb)_N per sample. Positive anomalies are highlighted in pink, negative in light blue.
- Trace-element ratios — 21 commonly used ratios per sample (La/Yb, Zr/Y, Th/Yb, Nb/Y, Ti/Y, Ti/V, Ba/Nb, Rb/Sr, …, and Nb/U and Ce/Pb, about constant in oceanic basalts and low in continental crust — Hofmann et al. 1986), with the Mg# and the alteration indices. Ti is the Ti column, or TiO2 × 5993.4 when there is none, as the diagrams take it. The same ratios are columns of the data table (section 6.6).
- Correlation matrix — Pearson r heatmap, poolable, facetable per group or shown for one group alone, with cell annotations if the matrix fits on screen. The trace ratios, alteration indices and anomalies are in the picker under their own headings, unticked so as not to swamp the matrix.
- Tectonic discrimination — 256 parameters, ticked from a tree and computed per sample. See below.
- Plots — the figures that go with the tables.

Mg#, Mg#_FeOt and, once the majors are adjusted, Mg#_adj are summarised like any oxide in the Descriptive, By group and Paper tables, and listed per sample with the trace ratios — and, beside the

data, in the data table (section 6.6); the saved report carries them in the same places. The Major elements panel under the toolbar decides whether the ratios and the tectonic parameters read adjusted or analysed majors.

The tectonic discrimination tab

This is where the discriminant functions behind the Verma diagrams, and the simple ratios behind the others, are reported as numbers. Tick what you want in the tree on the left; the count in the tab title follows your selection.

- For eight of the eleven Verma coefficient sets the posterior probability of each tectonic setting is given as well as the discriminant score, following Verma & Agrawal (2011). For Verma & Agrawal (2011) and Verma et al. (2012), whose papers do not print their centroids, they come from the authors' own Statistica sheet. The three earlier basic-rock sets (2004, 2006, 2008) classify by their drawn boundaries alone and report scores.
- 33.3 % is the by-chance level for a three-group panel. A sample sitting near it is inconclusive, and seeing the number says so where a field name would not.

Note. Every row here is checked against the diagram that plots the same quantity, and required to equal it exactly. The two reach the number by different code, so when a correction lands on one it has to land on the other.

12. Map and spatial analysis

Map → Sample locations and kriging… plots every sample at its longitude and latitude and, on the same axes, interpolates one numeric column across the area by ordinary kriging. Nothing is downloaded: there is no background map, and the view works offline.

12.1 Coordinates

Pick the format under Coordinate columns, then the columns it needs. B.I.R.D.S guesses them from common names (Latitude, Lat, Easting, Zone, …).

- Decimal degrees — 19.4326 and –99.1332; negative is south and west.
- UTM — easting and northing in metres, with the zone and hemisphere fixed for the whole dataset or read row by row from columns. The data template has the four columns, Easting, Northing, UTM_Zone and Hemisphere, beside Latitude and Longitude; fill whichever you have, and when only the UTM ones are filled the map says to pick UTM here.
- Degrees, minutes and seconds — strings such as 19°25'57"N.
- Decimal with a hemisphere letter — 19.42N, 99.13W.
- MGRS grid references — 14QMG5780012345 (needs the optional mgrs package).

Note. Every coordinate is taken as WGS-84. UTM is converted with pyproj when it is installed and with a built-in formula otherwise; the two agree to about a millimetre.

12.2 The sample map

The axes are longitude and latitude in degrees, with the aspect ratio fixed at the frame's centre latitude so that a square on the ground draws as a square there. This is the equirectangular projection, true along every meridian and along that parallel — as good as any projection over a field area. The scale bar is measured along that parallel on the WGS-84 ellipsoid.

- Auto-fit to samples frames the data with the padding you set. Untick it to keep a zoomed or panned frame through updates; the kriging grid then covers that frame.
- Markers take their colour and symbol from Data → Edit plot. Tick Colour by the analysed variable to colour them by value instead; samples without a value are drawn hollow.
- The scale bar, north arrow, grid and legend can each be turned off, and the annotation tools (lines, arrows, text, sample labels) work in degrees.

12.3 Kriging a variable

Choose the column under Spatial analysis → Variable and press Update map. The tabs above the plot switch between five displays:

- Samples — the locations, as above.
- Kriged estimate — the interpolated surface, with the samples over it as small white markers.
- Uncertainty — the kriging standard deviation. It is smallest at the samples and grows away from them; read it before trusting any part of the estimate.
- Variogram — semivariance against the distance between samples, each point sized by the number of pairs behind it, with the fitted model.
- Cross-validation — every sample against its value predicted from all the others. Points near the 1 : 1 line mean the spatial model predicts well.

The options:

- Transform — Logarithm (the default) kriges ln(value) and shows the median estimate. Concentrations are skewed and cannot be negative, which the logarithm respects; values at or below zero are left out and counted. Choose None for a variable that can be negative, such as εNd.
- Groups — Each group on its own (the default) kriges every group separately. A pluton and its host are not one continuous field, and kriging across the contact would draw a gradient that does not exist. Choose All groups pooled only when the groups are not separate bodies.
- Variogram model — spherical (the default), exponential or Gaussian. In all three the range is the effective range: the distance at which the model reaches 95 % of its sill.
- Lag classes and Maximum lag — how the experimental variogram is binned. Auto aims at about 30 pairs of samples per class, out to half the largest distance between two samples.
- Grid cells across — the resolution of the surface along the longer side of the frame.
- Clip to each group's sample hull — draw the surface only inside the outline of the samples, optionally with a buffer in km. Beyond it kriging can only extrapolate towards the mean.
- Contour lines on the estimate — labelled isolines over the surface.

Note. Distances are computed through the Earth between points on the WGS-84 ellipsoid, never read off the map, so no projection distorts them. Every group shares one colour scale, so colours compare between groups. Where two clipped surfaces overlap, the later group is drawn on top and the

summary says so.

12.4 Reading the summary

Below the plot a table gives, for each group, the samples used out of those available, the variogram model with its nugget, sill and range, and three cross-validation figures:

- Skill — the share of the error of simply guessing the mean that kriging removes. Near 1 the map adds a great deal; near 0 or below it adds nothing, whatever the surface looks like.
- RMSE — the typical prediction error, in the units kriged: natural logarithms under the default transform.
- MSSE — the mean squared standardised error. It is about 1 when the uncertainty map is honest; well above 1 it is too optimistic, well below 1 too cautious.

Notes under the table say in plain words what to doubt: too few samples for a reliable variogram (a variogram from fewer than 50 is of little value and about 100 are needed, after Webster & Oliver 1992); samples left out, or averaged because they share a location; a range that never levels off within the distances measured, which often means a regional trend; an ill-conditioned system; and overlapping surfaces. A group with fewer than five usable locations is not kriged, and the summary says why.

12.5 Setting the variogram by hand

- Pick the group under Variogram parameters → Group.
- Untick Fit automatically. The boxes start from the fitted values, so nothing moves until you change one.
- Type a nugget, sill or range and press Enter. The map, the variogram and the cross-validation all update, and the summary marks the model (by hand).
- Tick Fit automatically again to go back to fitting. Values set by hand belong to the variable and transform they were set for, and are dropped when either changes.

12.6 Exporting the grid to a GIS

Export grid for a GIS (.asc)… writes, for every kriged group, two ESRI ASCII grids in WGS-84 degrees — the estimate and its kriging standard deviation — each with a .prj file, and one text file recording the variable, the transform, and each group's model and cross-validation figures. Saved as Cu.asc with groups Pluton and Host, the files are:

```
Cu_Pluton.asc   Cu_Pluton_sd.asc   (and a .prj for each)
Cu_Host.asc     Cu_Host_sd.asc
Cu.txt
```

With a single surface — one group, or all pooled — the group is left out of the names. Cells outside the clipped hull hold the no-data value –9999. QGIS opens the grids with Layer → Add Layer → Add Raster Layer; ArcGIS opens them too, or converts them with its ASCII To Raster tool. The .prj places them without further setting.

Note. Kriging interpolates; it does not explain. A surface is only as good as the variogram behind it and the spread of the samples it was built from. Check the Uncertainty tab and the cross-validation skill before using a kriged map as evidence, and prefer the clipped surface in a figure.

13. Saving results

13.1 Save the statistics report

- Click Data → Export → Save statistics report…
- Choose .xlsx (recommended — one sheet per section) or .csv.
- Tick the sections you want (original data, descriptive, by-group, anomalies, trace ratios, correlation).
- Click Save.

13.2 Export a plot

- Make sure the plot you want is the one currently visible.
- Click Data → Export → Export plot… (or Copy plot to clipboard, to paste it straight into a document).
- Pick the format (.png, .jpg, .pdf, .svg) and the DPI (72–1200) and image dimensions in inches.
- Tick “Tight bounding box” to trim the margins and “Transparent background” for PNG / SVG figures meant for overlays.

13.3 Save and reopen a session

- Data → Save session (Ctrl+S) writes everything to one .birds file: the data sheets, the groups and their styles, each view's settings, and results already computed such as the CIPW norm, so reopening needs no recalculation.
- Data → Save session as… (Ctrl+Shift+S) writes it under a new name.
- Data → Open session… (Ctrl+O), or Data → Recent sessions, reopens one. With the application installed (sections 3.5 and 4.5) a double-click on the .birds file does the same.
- B.I.R.D.S also saves your work on its own every five minutes. If it closes before you save, the next start offers to recover it — unless it was started by double-clicking a session, which is then the one opened.
- Opening a session draws only the page it was saved on; the other views are brought up to date the first time you open them, so a session with many views opens in well under a second.

14. Troubleshooting

Opening a session or a dataset takes long

Google Drive, iCloud Drive, Dropbox and OneDrive can keep a file in the cloud only until something opens it; it then has to download first, which can take from a second to many minutes when the connection or the sync is slow. B.I.R.D.S reads the file in the background and shows “Downloading … from cloud storage” with a Cancel button: the window stays as it was, and Cancel leaves it untouched. To have your sessions open at once, keep the folder that holds them on the Mac: in

Finder, right-click the folder → Make available offline (Google Drive) or Keep Downloaded (iCloud).

B.I.R.D.S closed on its own

Reopen it: the work of the last five minutes is offered back. Then send the file ~/.birds/crash.log (in Finder, Go → Go to Folder… and type ~/.birds). Every Python error, every Qt warning and, for a crash inside Qt itself, the Python stack it happened under are written there. macOS keeps its own report as well, in Console → Crash Reports, under BIRDS.

“ModuleNotFoundError: No module named 'PySide6'”

The libraries are not installed yet. Run `pip install -r requirements.txt` from inside the BIRDS folder.

The CSV won't load

Open the file in a text editor and check that the first line contains the column headers, that there are no blank lines above them and that all rows have the same number of fields. If the file uses decimal commas (e.g. 50,5) B.I.R.D.S converts them automatically, but make sure the field separator is consistent (the same delimiter on every line).

A diagram shows the background but no samples

The diagram needs specific elements that are not present in your data. Look at the diagram's subtitle for the required columns and either rename your columns to match or add the missing measurements.

A normative diagram shows no samples

Barker, the haplogranite and Frost's FSSI panel need a CIPW norm for each sample, and the note in the corner of the diagram names what is missing. Most often the iron is given only as FeOt or Fe2O3t: split it in Data → Pre-process majors… and tick Use majors adjusted to 100 % (section 7, Diagrams drawn from the norm).

Group names don't match the legend

The legend label uses the name you typed in Data → Edit plot, matched to the i-th unique value of the grouping column. If the order looks wrong, reopen Edit plot and re-tick the column — names auto-refill in the order the values appear in the file.

Plot export looks pixelated

Increase the DPI in the Export plot dialog (300 dpi for submitted manuscripts, 600 dpi for camera-ready figures). For lossless quality prefer .pdf or .svg, which are vector formats.

The map says no samples could be parsed

The coordinate format or columns do not match the data. Check the format under Coordinate columns — UTM eastings read as decimal degrees fall far outside ±180° — and that latitude and longitude are not swapped.

The Variable list is empty

Only numeric columns with at least five values that are not all equal can be kriged, and the coordinate columns are left out. A column with a unit or a comment typed into its cells is read as text and has to be cleaned first. Values below detection (“<0.5”, “b.d.l.”) are not the problem: they are converted on import, under the policy in Data → Detection limits…

A group says “not kriged”

It has fewer than five usable locations, or every value is the same. Under the logarithmic transform, values at or below zero do not count. The other groups are kriged regardless.

“The fitted range ran to its upper bound”

The variogram keeps rising across every distance measured instead of levelling off. Often the variable follows a regional trend, and ordinary kriging, which assumes a constant mean, describes it only roughly. Try a smaller maximum lag, or read the map as a smoothed picture of the trend.

15. References

B.I.R.D.S implements diagrams and statistics from the following sources. Cite each one according to the diagram you used in your figures and tables — every plot prints its own citation as a subtitle, and References → Show full bibliography (APA) lists them all.

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